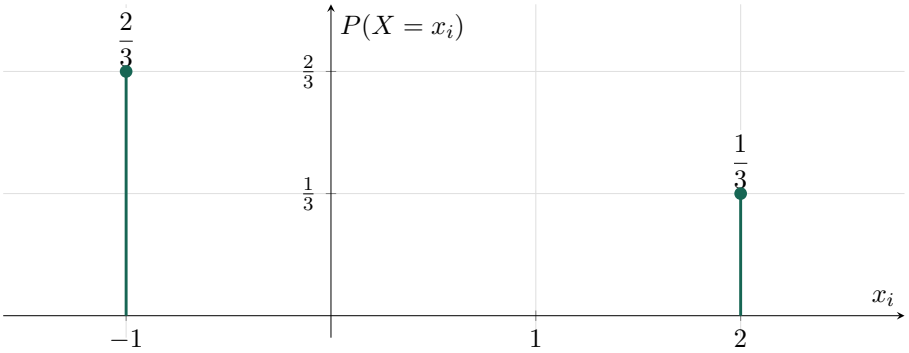
## 8.1 Variable, event and distribution

**Definition and results**

On a finite sample space ?, a real random variable X is a function that assigns a number real number to each outcome. The event $\{X = a\}$ groups the outcomes giving the value a. Its distribution assigns to each distinct value $x_i$ probability $p_i = P(X = x_i)$, with $p_i \geq 0$ and $\sum p_i = 1$.

**Status and justification.** Definitions. The events associated with distinct values form a partition of the sample space; this justifies adding the probabilities.

**Worked example.** A fair die gives a winning X equal to 2 for 1 or 2, and to $-1$ otherwise. The distribution is $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$. $\{X \leq 0\} = \{3; 4; 5; 6\}$ : probability $2\frac{-}{3}$.




The height indicates the probability: $P(X = -1) = \frac{2}{3}$ and $P(X = 2) = \frac{1}{3}$. The exact probabilities appear on both bars.

**Watch out**

The values are not always equally likely even if the outcomes are. Group the outcomes which give the same gain.

**Immediate application.** Two fair coins are tossed. X counts the tails. Write its distribution and $P(X \leq 1)$.

## 8.2 Expectation and linearity

**Definition and results**

The expectation is $E(X) = \sum_i x_i p_i$. It describes the theoretical long-run mean, not an outcome necessarily possible on a single trial. For variables X and Y on the same finite sample space and real numbers a,b: $E(aX + b) = aE(X) + b$ and $E(X + Y) = E(X) + E(Y)$, without assumption of independence.

**Status and justification.** Summing over the outcomes $\omega$, $E(X + Y) = \sum_\omega (X(\omega) + Y(\omega))P(\omega)$ ; distributing the sum gives the result. The constant b has expectation b because the probabilities sum to 1.

**Worked example.** For the previous distribution $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$, $E(X) = 0$ : the game is fair if X is the net gain. For $Y = 3X - 2$, $E(Y) = 3 \times 0 - 2 = -2$.

**Watch out**

An expectation is not an individual guarantee; a game can be a losing one over any horizon observed despite a positive expectation.

**Immediate application.** A gross winning is 10 with probability $\frac{1}{5}$ and 0 otherwise. What bet makes the fair game?

## 8.3 Variance, standard deviation and König·Huygens

**Definition and results**

Denoting $\mu = E(X)$, $V(X) = E((X - \mu)^2) = \sum_i (x_i - \mu)^2 p_i$. The standard deviation is $\sigma(X) = \sqrt{V(X)}$. The formula of König·Huygens gives $V(X) = E(X^2) - E(X)^2$. A variance is always positive or zero; the standard deviation has the same unit as X, the variance a squared unit.

**Status and justification.** Expand ( $X - \mu)^2 = X^2 - 2\mu X + \mu^2$, apply linearity and $E(X) = \mu$ : $E(X^2) - 2\mu^2 + \mu^2$. We get the formula.

**Worked example.** For X with values 0 and 4 with probabilities $\frac{3}{4}$ and $\frac{1}{4}$, $E(X) = 1$, $E(X^2) = 4$, $V(X) = 3$ and $\sigma = \sqrt{3}$. Direct calculation: $(0 - 1)^2(\frac{3}{4}) + (4 - 1)^2(\frac{1}{4}) = 3$.

**Watch out**

$E(X^2)$ is not $E(X)^2$. A negative computed variance signals an error or a very small rounding error to be dealt with explicitly.

**Immediate application.** For X with values $-1$, 1 equally likely, calculate the expectation, variance and standard deviation.

## 8.4 Transforming and comparing gains

**Definition and results**

For real a, b, $V(aX+b) = a^2V(X)$ and $\sigma(aX+b) = |a|\sigma(X)$. A translation moves the mean without changing the spread; a multiplication changes the scale. Two data sets with the same expectation can have very different spreads.

**Status and justification.** $aX+b-E(aX+b) = a(X-E(X))$. Squaring and then taking the expectation, we get $a^2V(X)$. The square root of $a^2$ is $|a|$.

**Worked example.** Game A always pays 2. Game B pays 0 or 4 with equal probability. The two have expectation 2; variances 0 and 4. Their distributions are therefore not interchangeable despite the same mean.

**Watch out**

The variance of a sum is not, in general, the sum of the variances; no rule of this deviation is used without hypotheses.

**Immediate application.** $E(X) = 3$, $V(X)$ = 4. Give the parameters of $Y = -2X + 5$.

## 8.5 Samples and the observed mean

**Definition and results**

A sample of size n is here a list of n independent realisations of the same variable X. Its observed mean m varies from one sample to another. We compare m with $\mu = E(X)$; some repetitions make it possible to observe the proportion of samples satisfying $|m-\mu| \leq \frac{2\sigma}{\sqrt{n}}$. This threshold is an experimental task in the curriculum, not a claim about coverage rates exact.

**Status and justification.** Experimentation: no convergence theorem or law is proved here normal. We check the size of the lists, the calculation of the mean and the distinction between n and the number N samples.

**Worked example.** For X with equally likely values 0 and 2, $\mu = 1$, $\sigma$ = 1. For n=100, threshold $\frac{2}{10} = 0{,}2$ : count the means in [0 ,8; 1,2]. A result of 956 out of 1000 means a frequency observed 0.956 for this simulation only.

**Watch out**

Do not copy a single draw n times; do not state that the proportion is necessarily 95 %.

**Immediate application.** For $\mu = 5$, $\sigma$ = 3 and n=36, give the requested experimental interval.